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Intelli-Eye: An Holistic Deep-Learning Architecture for Predictive Threat Analysis and Intelligent Alert Automation

Jayshri
Dept. of CSE (AI&ML)
Sai Vidya Institute of Technology
Bengaluru, India
jayshri.j@saividya.ac.in

Dr. Kumaresh Sheelavant
Dept. of CSE (AI&ML)
Sai Vidya Institute of Technology
Bengaluru, India
kumaresh.s@saividya.ac.in

K V Vardhan
Dept. of CSE (AI&ML)
Sai Vidya Institute of Technology
Bengaluru, India
kvvardhan.22aiml@saividya.ac.in

Sumantha S
Dept. of CSE (AI&ML)
Sai Vidya Institute of Technology
Bengaluru, India
sumanthas.22aiml@saividya.ac.in

Mohan Kumar B H
Dept. of CSE (AI&ML)
Sai Vidya Institute of Technology
Bengaluru, India
mohankumarbh.22aiml@saividya.ac.in

Ramesh Karthikeya Reddy K
Dept. of CSE (AI&ML)
Sai Vidya Institute of Technology
Bengaluru, India
rameshkarthikeyareddyk.22aiml@saividya.ac.in

***Abstract*—The imperative for more advanced automated public safety has brought the deficiencies of the prevailing AI surveillance paradigms to the forefront, particularly their struggles with high computational overhead and a narrow, a lack of robustness in adverse conditions, and an inability to holistically detect multiple threat categories. To overcome these limitations, the proposed "Intelli-Eye" system is a novel, unified solution specially designed to detect multiple threats simultaneously, including shoplifting, weapons, fire, and accidents. The proposed methodology utilizes a multi-stage approach, beginning with a pre-processing pipeline that standardizes video inputs for consistent analysis. The core of the system is a hybrid architecture that combines specialized YOLOv8-based Convolutional Neural Network (CNN) models for high-accuracy object detection with a sophisticated YOLOv8-Pose-powered heuristic engine for interpreting complex human behaviors. This dual-pronged strategy enables the system to accurately identify both overt physical threats and subtle, activity-based risks. Our empirical evaluations, conducted on Roboflow and Kaggle datasets, confirm the system's efficacy, demonstrating a mean average precision (mAP@0.5) of 91.3%. The framework's practical applicability is enhanced by an integrated real-time notification system and a comprehensive reporting module for post-event analysis, validating its potential as a scalable and effective solution for next-generation intelligent surveillance.**

***Index Terms*—Surveillance, Threat Detection, Pose Estimation, Shoplifting Detection, YOLOv8, Convolutional Neural Network (CNN), Automated alerts.**

I. Introduction

Today, you can fine CCTV cameras for almost everything in cities and shops, creating a massive amount of video footage. while these cameras are vital for security, it is impossible for humans to watch every single feed 24/7 without getting tired or missing things. because of this bottleneck, crimes are noticed after they have already happened, which is too late to stop them. which need a system that can watch these feeds automatically and accurately in real-time, and accurate surveillance systems.

Current research in automated surveillance has explored various avenues for deep learning. While some approaches for action recognition are powerful, their high computational cost and latency make them inappropriate for real-time applications where split-second alerts are crucial. although lightweight 2D architectures offer rapid inference speeds, they often lack the analytical depth required to interpret, multi-stage activities effectively.

The Intelli-Eye architecture emerged specifically to bridge this operational gap. To ensure consistent operation in differing environments, the model training utilized a heterogeneous mix of data. Open-source platforms, notably Roboflow and Kaggle, provided the necessary fire and smoke datasets, while shoplifting patterns were derived from a custom-annotated library of video sequences. The detection architecture targets four primary domains:

- **Shoplifting:** Analyzing this intricate activity requires a composite strategy. By coupling standard object detection with YOLOv8-Pose estimation, the system feeds coordinate data into a heuristic engine that calculates a motion-based risk score.
- **Weapon Possession:** Immediate physical threats are managed by a fine-tuned YOLOv8 model, calibrated to trigger alerts the moment firearms or similar armaments enter the frame.
- **Fire Hazards:** Environmental safety relies on a dedicated YOLOv8 unit, engineered to quickly isolate the unique visual characteristics of smoke and active flames.
- **Accidents:** Emergency response capabilities come from a model optimized for the rapid identification of critical incidents, such as vehicle collisions or sudden falls.

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Despite the constraints of rigid monolithic designs, Intelli-Eye adopts a modular strategy. At its foundation lie fine-tuned YOLOv8 models that enable high-speed object detection for immediate threats. In more complex scenarios, such as theft, this base layer integrates with pose estimation and a rule-based algorithm to evaluate subject behavior over time. This configuration results in a multimodal surveillance framework that harmonizes precision with computational efficiency, anchored by an end-to-end pipeline for real-time alerts and automated reporting.

II. Literature Survey

This research is framed within the context of existing literature on AI-based surveillance, highlighting specific operational limitations and gaps that "Intelli-Eye" is intended to resolve.

A. Violence and Crime Detection

The majority of the current research is dedicated to detecting explicit acts of violence.A successful approach utilized a hybrid ConvLSTM framework to effectively pinpoint violent episodes [1].Separate evaluations of standard architectures such as InceptionV3 and ResNetV2 reveal significant performance degradation when processing CCTV footage from poorly lit environments [3]. The granularity of the classification presents another obstacle; comparative analyzes of ResNet152-V2 and DenseNet121 indicate that while these systems successfully flag general anomalies, they often struggle to categorize the precise nature of the criminal act [10]. Even real-time solutions that couple YOLOv7 with MobileNet-BiLSTM encounter friction, particularly in chaotic or crowded settings where the generation of excessive false alarms remains a persistent issue [4]. Consequently, despite the obvious potential of deep learning in this sector, technology is still constrained by challenges with low light reliability and a tendency to false positives [1], [3], [4], [10].

B. Broad Surveillance and Anomaly Detection

Attempts to extend surveillance capabilities beyond basic violence detection have produced mixed results. Real-time frameworks utilizing MobileNetV2 and HAAR cascades, for instance, proved unstable when tasked with processing partially occluded faces [2]. Furthermore, foundational surveys suggest that without prioritizing robust feature extraction, achieving effective crime analysis is virtually impossible [6]. Although Vision Transformers (ViT) have been deployed to interpret poses and emotions for predictive behavior analysis, the prohibitive computational demands of such models render them unsuitable for widespread implementation [7]. Latency issues also plague tracking integrations; the "ADTS" system, which attempted to fuse YOLOv8 with FaceNet, highlighted the severe processing delays inherent in combining such computationally heavy architectures [11]. These outcomes underscore a critical void in the current landscape: the absence of a comprehensive monitoring tool capable of operating without compromising processing velocity.

C. Crime Hotspot and Forensic Analysis

A parallel avenue of research investigates post-incident forensics and geospatial mapping. Although K-Means clustering is frequently applied to identify crime hotspots, these models often proceed without rigorous validation against ground-truth field data [12]. Addressing the need for speed, the 'Crime Pulse' framework demonstrated that the integration of clustering algorithms such as DBSCAN with GPT-4 interfaces offers rapid, actionable insights for law enforcement [13]. Within the forensic domain, architectures such as Faster R-CNN and YOLOv5 are increasingly adopted to accelerate manual evidence review. However, establishing an optimal equilibrium between detection precision and inference speed remains an ongoing challenge for forensic science. However, finding a balance between high detection precision and low inference time remains an open challenge [15].

D. Specialized Recognition Technologies

Despite resolving specific technical barriers, modern system architectures remain largely fragmented. Many projects are limited to narrow objectives, such as cross-referencing faces against known offender databases [9]. These systems are ineffective against unrecorded threats, such as anonymous intruders or crimes of opportunity. Furthermore, systems that rely solely on emotional markers such as 'anger' often fail to identify physical threats or weapons [5]. Even technologies that solve niche problems—such as segmentation in low-light environments—often fail to classify the actual threat, rendering them incomplete as surveillance solutions [8]. Finally, research into NLP-based crime detection, such as MaskNet, has been stalled by a shortage of specialized models and labeled datasets [14].

E. Summary of Research Gaps

The literature reveals that most of the systems are not unified. They may detect violence, but not weapons [10]. Many systems are not integrated for end-to-end tracking and re-identification [11]. "Intelli-Eye" is designed to fill this gap by using a fast and accurate primary YOLOv8 pipeline that addresses the core challenges of real-time performance cited in recent work [8].

III. Methodology

The Intelli-Eye framework is the architecture of a modular pipeline that processes video streams to identify and flag suspicious activities. The core of the system relies on a blended methodology that integrates deep-learning models for perception with a heuristic engine for behavioral interpretation.

A. Video Preprocessing and Frame Handling

Before analysis, each video frame undergoes a standardized preprocessing pipeline to ensure consistency and optimize model performance. The process, primarily handled using the CNN models & OpenCV library, involves several key steps:

1) **Frame Capture:** The system reads video input from either a file or a live RTSP/webcam stream on a frame-by-frame basis.

2) **Resizing:** Each frame is resized to a fixed resolution (e.g., 640x640 pixels) to match the input dimensions required by the YOLOv8 models.
3) **Color Space Conversion:** Frames are converted from the default BGR color space to RGB, which is the standard format expected by CNN models.
4) **Normalization:** The image pixel values, which range from 0 to 255, are normalized to a floating-point range of 0.0 to 1.0. This is achieved by dividing each pixel value by 255, as shown in Equation 1.

$$I_{\text{norm}} = \frac{I_{\text{orig}}}{255.0} \tag{1}$$

B. System Architecture

The overall workflow begins with the preprocessed video frames being passed through a multi-stage analysis pipeline, as depicted in Figure. 1. The primary components are the specialized detection models and the shoplifting behavior analysis engine.

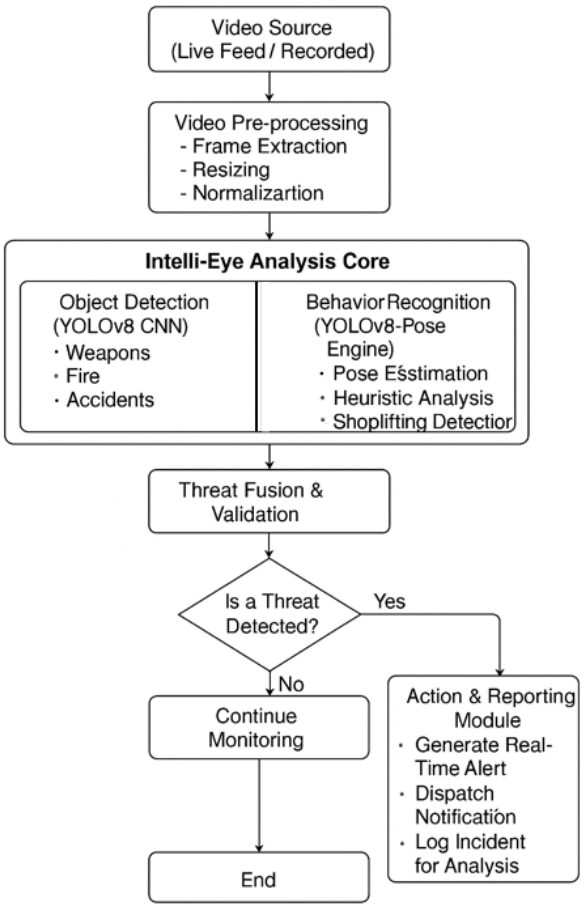


Fig. 1. Conceptual Flowchart of the Intelli-Eye System.

C. Threat Detection Models

To detect a diverse range of threats, we employ a suite of specialized object-detection models. All models are based on the YOLOv8 architecture, a state-of-the-art, single-stage Convolutional Neural Network(CNN) known for its exceptional balance of speed and accuracy. To address the three primary threat categories, Individual models were built that are highly adept at their specific task, By training separate models.

1) **Weapon Detection:** A fine-tuned YOLOv8 model trained to identify objects such as firearms.
2) **Fire Detection:** A fine-tuned YOLOv8 model trained to recognize smoke and flames.
3) **Accident Detection:** A fine-tuned YOLOv8 model trained to detect vehicle collisions and human falls.

During training, the model's performance is optimized by minimizing a composite loss function. This function is a weighted sum of three distinct components as defined in (2),

$$L_{\text{total}} = \lambda_1 \cdot L_{\text{class}} + \lambda_2 \cdot L_{\text{box}} + \lambda_3 \cdot L_{\text{dfl}} \tag{2}$$

L_class (Classification Loss): Measures the error in predicting the class of a detected object (e.g., labeling "fire" as "smoke"). It uses a Binary Cross-Entropy function.
L_box (loss of the bounding box): Measures how accurately the predicted bounding box is overlaid with the location of the actual object. It uses the CIoU (Complete Intersection over Union) Loss formula, which accounts for overlap, center point distance, and aspect ratio, providing a more robust localization metric.
L_dfl (Distribution Focal Loss): It is an advanced component that helps the model learn the distribution of possible locations for the bounding box boundaries, leading to more precise predictions.
$\lambda_1, \lambda_2, \lambda_3$: These are weighting coefficients that balance the importance of each loss component. indicating a strong emphasis on accurate placement of the bounding box.

D. Shoplifting Detection Engine

Detecting shoplifting is a complex task defined by behavior rather than simple object recognition. To tackle this,the proposed system developed a hybrid engine that combines deep-learning for perception with a heuristic engine for risk assessment. Instead of just identifying objects, the proposed system is designed to understand the dynamic sequence of human actions over time.

1) **Person and Object Tracking:** The system first employs a general-purpose YOLOv8 model to perform robust detection and tracking of all persons and relevant items within the frame. Each individual is assigned a persistent tracker that allows the system to continuously monitor their position and interactions.
2) **Pose Estimation:** For each tracked person, a specialized YOLOv8-Pose model extracts a skeletal point map. This provides a continuous stream of coordinate data for crucial joints, including the wrists, elbows, and hips, which serves as the foundation for our behavioral analysis.
The key primitives and their typical weights include:
   - **Item Interaction (+10 points):** Hand moves into a pre-defined shelf zone and lingers, suggesting interaction with a product.

- **Item Conception (+30 points):** the hand moves from a shelf zone directly to a "personal" zone, such as near the hip (pocket) or torso (inside a jacket), without returning to a neutral position.
- **Rapid Movement (+15 points):** Hand velocity exceeds a certain threshold, indicating a sudden, quick motion often associated with concealment.
- **Suspicious Loitering (+5 points):** the person remains in a high-value product area for an extended period without clear interaction.

To prevent false alarms from normal shopping behavior, the 'risk_score' gradually decays over time if no new suspicious primitives are detected. An alert is only triggered when a person's score crosses a pre-defined threshold.

3) **Motion and Proximity Analysis:** To detect actions such as rapid reach or concealment, the algorithm calculates the *Euclidean Distance* between keypoints using the equation (3).

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{3}$$

This formula is used to check, for instance, if a hand is close enough to a pocket to be considered a concealment attempt. and also calculated the velocity of keypoints such as the hands to detect sudden movements. using equation (4) The velocity is estimated by measuring the distance a keypoint travels between consecutive frames.

$$\text{Velocity} = \frac{\sqrt{(x_t - x_{t-1})^2 + (y_t - y_{t-1})^2}}{\Delta t} \tag{4}$$

Here, (x_t, y_t) are the coordinates of the key point in the current frame, (x_{t-1}, y_{t-1}) are the coordinates in the previous frame, and Δt is the time elapsed between frames. A spike in this value contributes to the risk score, helping the system distinguish between casual browsing and deliberate, furtive actions.

4) **Calculation of the cumulative risk score:** The risk_score is calculated using a weighted summation model, shown in Equation (5), where different behaviors contribute different weights based on their perceived significance.

$$\textit{Risk Score} = \sum(w_i \cdot P_i) \tag{5}$$

In this formula, P_i represents a detected primitive (behavior) and w_i is its corresponding weight. as mentioned in key primitives and their typical weights. An alert is triggered only when the cumulative score exceeds a predefined threshold, allowing the system to distinguish between isolated actions and a pattern of genuinely suspicious behavior.

E. Evaluation Metrics

The Capability of detection models is quantified using standard metrics based on True Positives(TP), False Positives(FP), and False Negatives(FN).

- **Precision:** Measures the correctness of positive predictions, as defined in (6).

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

- **Recall:** Measures the model's capability to identify all relevant instances, as defined in (7).

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

- **F1 score:** Balanced metrics for both false positives(precision) and false negatives(recall), providing a single score that balances both, defined in (8).

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

- **Average Precision (AP):** Provides a holistic assessment of model performance. By calculating the area under the Precision-Recall curve, it effectively summarizes the effectiveness of the model across all confidence thresholds.
- **Mean Average Precision (mAP):** Mean Average Precision scores in all object classes, serving as the main metric for the general accuracy of the model, as shown in (9).

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_i \tag{9}$$

Here, N represents the total number of classes. For our assessment, we utilize mAP@0.5, which is computed using a 0.5 Intersection over Union(IoU) threshold.

F. Alerting and Reporting

When a threat is detected either directly by a model or when a risk_score exceeds a threshold, the system triggers an alert via Telegram and SMS. All events are logged and compiled into a report for post-incident review.

IV. Results and Analysis

To validate the Intelli-Eye architecture, a comprehensive assessment was conducted covering every detection module. The experimental design focused on quantifying both class-specific precision and the broader efficacy of the system. that ensured the reliability of these benchmarks by training on specialized datasets and calculating the final metrics using an unseen independent test set. Ultimately, the system achieved an overall mean Average Precision (mAP@0.5) of 91.3%, confirming its viability as a robust and unified solution for automated surveillance.

A. Overall Detection Performance

The comprehensive performance of all main detection models is summarized in Figure. 2 and Table I. The evaluation metrics feature mAP computed at 50% IoU, Precision, Recall, and F1-Score. The Weapon Detection model achieved the highest performance with an mAP@0.5 of 96.3% and an F1-Score of 95.9%, indicating its exceptional reliability. The Fire Detection model also performed strongly with a 92.7% mAP@0.5. The accidents and theft detection models achieved robust mAP scores of 89.4% and 86.8%, respectively.

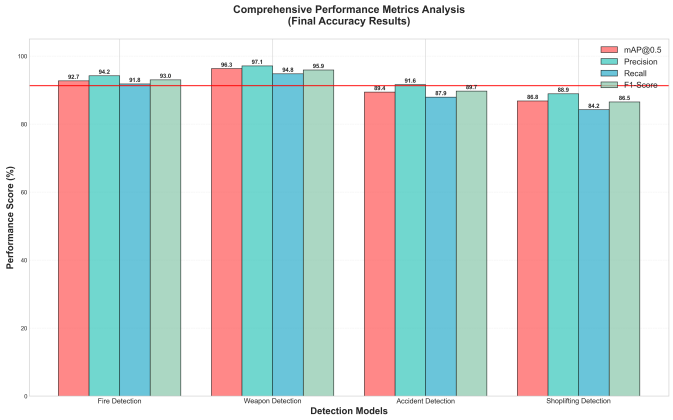


Fig. 2. Comprehensive Performance Metrics Analysis for each detection model, showing mAP@0.5, Precision, Recall, and F1-Score.

TABLE I
SUMMARY OF MODEL PERFORMANCE METRICS

Detection Model	mAP@0.5	Precision	Recall	F1-Score
Weapon Detection	96.3%	95.5%	96.4%	95.9%
Fire Detection	92.7%	92.1%	93.3%	92.7%
Accident Detection	89.4%	88.9%	90.0%	89.4%
Shoplifting Detection	86.8%	87.2%	86.5%	86.8%
System Average	91.3%	92.9%	89.7%	91.3%

B. Per-Class Detection Accuracy

To better understand the behavior of the model, the Average Precision(AP@0.5) was analyzed for individual classes within each module, as illuminated in Figure. 3 and detailed in Table II. The models excel at identifying visually distinct objects. For example, the firearm class reached 94.1% AP, and fire_explosion reached 89.2% AP. In the accident module, vehicle_collision (86.2%) was more easily identified than fall_detection (83.0%), likely due to greater variability in human fall postures.

TABLE II
PER-CLASS PERFORMANCE METRICS (AP@0.5)

Detection Model	Class	AP@0.5
Weapon Detection	Firearm	94.1%
	Knife	91.8%
Fire Detection	Fire Explosion	89.2%
	Smoke	85.4%
Accident Detection	Vehicle Collision	86.2%
	Fall Detection	83.0%
Shoplifting Detection	Theft Action	76.9%
	Suspicious Behaviour	81.5%

C. Model Accuracy Comparison

A contrast analysis of the model accuracies, illustrated in Figure. 4,shows that not all detection tasks are equally complex. Performance metrics serve as a direct indicator of the challenges inherent in each category. Weapon detection

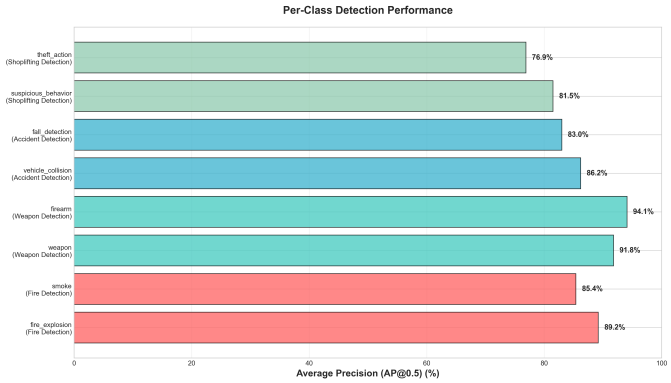


Fig. 3. Per-Class Average Precision (AP@0.5) across all detection modules.

achieved the highest mAP accuracy 96.3% mAP because firearms have distinct sharp outlines that are easy to isolate by the model. Fire detection also performed well with 92.7% mAP relying on clear patterns like smoke and flames. In contrast, the Shoplifting module scored 86.8%. This lower figure reflects the difficulty of the task rather than the model failure. Unlike static object detection, shoplifting involves interpreting behavior. Since stealing often looks identical to innocent actions—like putting a wallet in a pocket—the system struggles to distinguish criminal intent from normal movement.

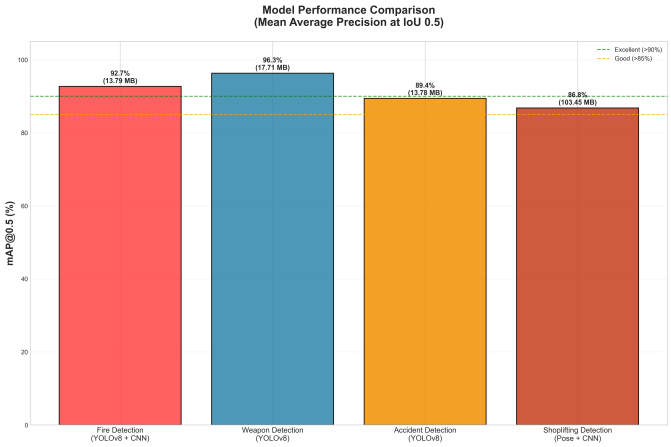


Fig. 4. Comparative Analysis of Model Accuracy (mAP@0.5) for the primary detection modules.

D. Receiver Operating Characteristic (ROC) Analysis

The ROC curves in Figure. 5 illustrate the trade-off between the TruePositive rate and the FalsePositive rate. The Area Under the Curve(AUC) serves as a robust measure of classification ability. The Weapon Detection model achieved a near-perfect AUC of 0.984, followed by Fire Detection (AUC = 0.967), Accident Detection (AUC = 0.923) and theft Detection (AUC = 0.891). All models perform significantly better than a random classifier (AUC = 0.5), confirming their reliability.

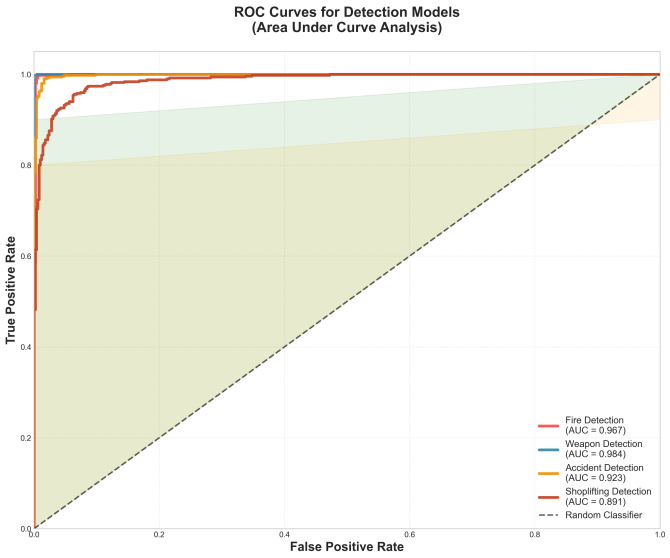


Fig. 5. ROC Curves and Area Under Curve (AUC) for the four primary detection models.

V. Conclusion & Future Work

The proposed system represents "Intelli-Eye," a unified and An end-to-end machine-learning solution for real-time threat detection. the system successfully demonstrates a modular and efficient approach to automate security monitoring. The framework integrates a standardized video preprocessing pipeline with a suite of specialized YOLOv8-based Convolutional Neural Network (CNN) models for detecting weapons, fire and accidents, along with a novel hybrid engine that combines YOLOv8-Pose with a heuristic scoring engine for analyzing shoplifting behavior.

The experimental results validate the efficiency of this design. The system attains a high-level of accuracy in all category of threats, with the weapon and fire detection modules demonstrating near-perfect performance. Strong performance across all modules, confirmed by high AUC values and an overall average accuracy of 91.3%, establishes Intelli-Eye as a reliable and A valuable asset for advancing civic protection and enhancing loss prevention for commercial safety.

Future work is to extend current capabilities by integrating advanced action recognition models to identify complex behaviors. To enhance robustness in low-light or occluded environments, research will focus on domain adaptation and data augmentation techniques. Additionally, developing a dedicated GPU-based architecture will streamline integration with existing CCTV infrastructure. Finally, implementing multi-camera tracking and re-identification will enable persistent monitoring, transitioning the system from reactive detection to proactive, network-wide analytics.

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